

title: Hummingbird Lab 3 - PD Control (Simulation) date: 2025-10-31 header-
includes:

- \usepackage
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- \graphicspath{{.../figs/}{figs/}{_hummingbird_sim/figs/}} resource-path:
- .
- _hummingbird_sim
- figs
- .../figs

Overview

- This note summarizes the PD designs used in Lab 3 for the hummingbird simulation and records the specific formulas used to compute gains from time-domain specs. All math is LaTeX to render correctly when exported to PDF (tested with pandoc and typical VS Code PDF exporters that support MathJax/KaTeX).

Models

- Longitudinal (about $\theta = 0$):

$$\theta' = \frac{\theta}{\frac{m_{\ell_2} + m_{\ell_2'} + J}{F} + \frac{m_{\ell_2} + m_{\ell_2'} + J}{F} + \frac{m_{\ell_2} + m_{\ell_2'} + J}{F}}$$

- Lateral (successive loop closure, inner roll faster than outer yaw):

$$\begin{aligned} \phi_{//}(t) &= \frac{1}{J} \tau(t), \\ \psi_{//}(t) &\approx b \phi(t), b \approx \ell \frac{F}{J}, \\ J_{zz,eq} &= m \frac{\ell_1}{2} + m \frac{\ell_2}{2} + J_{2z}^{\psi} + J_{1z}^{\psi} + \frac{J_{3,eq}^T}{m} (\ell_{3x}^2 + \ell_{3y}^2) + J_{3z}. \end{aligned}$$

Dirty Derivative (First-Order Filtered Difference)

For any signal x :

$$\dot{x}_\alpha[k] = \alpha \dot{x}[k-1] + (1-\alpha) \frac{x[k] - x[k-1]}{T_s},$$

$$\alpha = \frac{\tau}{\tau + dT_s}$$

Gain Mapping from Specs

With desired ζ and rise time t_r :

$$\omega_n \approx 2.2 \cdot \frac{1}{t_r}$$

- Longitudinal (pitch):

$$k_{P_\theta} = \frac{\omega_n^2}{g n_\theta}, k_{D_\theta} = \frac{2\zeta \omega_n}{g n_\theta}$$

- Inner roll:

$$k_{P_\phi} = \frac{\omega_n^2}{n_\phi J_{xx}}, k_{D_\phi} = \frac{2\zeta \omega_n}{n_\phi J_{xx}}$$

- Outer yaw:

$$k_{P_\psi} = \frac{\omega_n^2}{g n_\psi}, k_{D_\psi} = \frac{2\zeta \omega_n}{g n_\psi}$$

Bandwidth Separation

Choose inner-loop roll at least one order faster than yaw:

$$\omega_{n_\phi} = M \omega_{n_\psi}, M \in [10, 20].$$

Control Laws (PD Only)

- Longitudinal PD (force):

$$\begin{aligned} e_\theta &= \theta_d - \theta, \quad \dot{e}_\theta = -\dot{\theta} \\ \tilde{F} &= k_{P_\theta} e_\theta + k_{D_\theta} \dot{e}_\theta, \quad F = F_e + \tilde{F} \end{aligned}$$

- Lateral PD (successive loops):

Outer yaw (commands roll reference):

$$e_{\psi} = \psi_d - \psi, \quad \widehat{\dot{e}}_{\psi} = -\widehat{\dot{\psi}}, \quad \phi_d = k_{P_{\psi}} e_{\psi} + k_{D_{\psi}} \widehat{\dot{e}}_{\psi}.$$

Inner roll (commands torque):

$$e_{\phi} = \phi_d - \phi, \quad \widehat{\dot{e}}_{\phi} = -\widehat{\dot{\phi}}, \quad \tau = k_{P_{\phi}} e_{\phi} + k_{D_{\phi}} \widehat{\dot{e}}_{\phi}.$$

Simulation Notes

- H.7: Step θ by $\pm 5^\circ$ and verify well-damped tracking. The logged force F should equal F plus PD action.
- H.8: Step ψ (e.g., 30°). The roll response should be visibly faster; the rise-time ratio should be near M .

How to Run

From the repository root (recommended to ensure imports resolve):

```
python -m _hummingbird_sim.h7_hummingbirdSim
python -m _hummingbird_sim.h8_hummingbirdSim
```

Or from the directory itself (matches current import style):

```
cd _hummingbird_sim
python h7_hummingbirdSim.py
python h8_hummingbirdSim.py
```

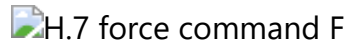
Python version used during development: 3.13. If matplotlib/pyqt are missing, install from the repo root:

```
python -m pip install --user -r requirements.txt
```

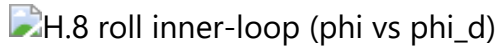
Results and Figures (placeholders)

GUI Simulation Output from H.7 & H.8 Images are expected in `../figs/` relative to this Markdown file (as you currently have in the repo). The YAML header sets a LaTeX `\graphicspath` and a Pandoc resource-path so PDF export can find them even if you run Pandoc from the repo root.

H.7 - Longitudinal PD (Pitch)

 H.7 pitch tracking (theta vs theta_d)  H.7 force command F

H.8 - Lateral PD (Yaw/Roll)

 H.8 yaw tracking (psi vs psi_d)  H.8 roll inner-loop (phi vs phi_d)

Tip: In matplotlib windows you can use the save icon, or temporarily add `plt.savefig('figs/h7_theta.png', dpi=200, bbox_inches='tight')` near the end of the script.

PDF Export

Pandoc (recommended). Run from the file Ts folder so relative image paths like `../figs/...` resolve the same as VS Code preview:

```
cd _hummingbird_sim
pandoc Lab3_Review.md \
  --standalone \
  --from gfm+tex_math_dollars+attributes+link_attributes \
  --to pdf \
  --pdf-engine=lualatex \
  -o Lab3_Review.pdf
```

Alternative (run from repo root) ?" include an explicit resource path so Pandoc finds figures stored outside this folder:

```
pandoc _hummingbird_sim/Lab3_Review.md \
  --standalone \
  --from gfm+tex_math_dollars+attributes+link_attributes \
  --to pdf \
  --pdf-engine=lualatex \
  --resource-path=".;_hummingbird_sim;figs;_hummingbird_sim/figs" \
  -o _hummingbird_sim/Lab3_Review.pdf
```

VS Code: pick a PDF exporter with MathJax/KaTeX enabled. If formulas render as plain text or images are missing, switch to Pandoc or use the `cd _hummingbird_sim` approach above.

Repo Map (for this lab)

- Controllers: `ctrlLonPD.py`, `ctrlPD.py`
- Sims/plots: `h7_hummingbirdSim.py`, `h8_hummingbirdSim.py`, `dataPlotter.py`

- Dynamics/params: hummingbirdDynamics.py, hummingbirdParam.py, signalGenerator.py
- This report: Lab3_Review.md, lessons: Lab3_Learned.txt

Troubleshooting

- If you see OpenGL/pyqtgraph warnings during animation, they are typically harmless on headless or constrained GPUs.
- If `_hummingbird_sim` imports fail when using `-m`, run from inside the folder with `python h7_hummingbirdSim.py`.